

Nvidia Corp. (NVDA)

Clarity on Blackwell and Strong Data Center Demand Met with Margin Re-Set; FY2Q EPS Recap

Buy

NVDA	12m Price Target: \$135.00	Price: \$125.61	Upside: 7.5%
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While we expect Nvidia's FY4Q (January) implied gross margin outlook in the low-70% range coupled with the company's growing opex budget to drive a re-set in out-year Street margin expectations, we remain constructive on Nvidia's Data Center revenue opportunity that spans Cloud, Consumer Internet and Enterprise customers and Training and Inference workloads. Positively, on Blackwell (i.e. next-generation Compute platform), management confirmed that they had executed a re-design without any compromise on performance, and that it could contribute several billion dollars in revenue in FY4Q on top of what is still a growing Hopper revenue stream. In our model, we reduce FY2026/27 non-GAAP gross margin (excl. SBC) forecasts by 200bps/210bps, but modestly increase our FY2026/27 non-GAAP EPS (excl. SBC) estimates by 3%/1%, primarily on higher Data Center revenue. We maintain our Buy rating (also on the Americas Conviction List) as well as our 12-month price target of \$135. See our full note for more details.

Results and Guidance

Results: FY2Q (July) revenue of \$30.0bn increased 15% qoq and 122% yoy. By segment, **Gaming** revenue of \$2.9bn was up 9% yoy driven by solid back-to-school demand. **Data Center** revenue of \$26.3bn was up 16% qoq reflecting strong demand for the Hopper GPU computing platform as well as continued growth in InfiniBand and a doubling of Ethernet for AI revenue. **Professional Visualization** revenue of \$454mn increased 6% qoq with growth driven by the continued ramp of RTX GPU workstations. **Automotive** revenue of \$346mn was up 5% qoq, driven by sequential growth in AI Cockpit solutions and self-driving platforms. **OEM and Other** revenue of \$88mn increased 13% qoq. Non-GAAP gross margin (excl. SBC) of 75.7% was down 320bps qoq due to inventory provisions for low-yielding Blackwell material and a higher

Toshiya Hari

+1(646)446-1759 | toshiya.hari@gs.com
Goldman Sachs & Co. LLC

Anmol Makkar

+1(212)934-5814 | anmol.makkar@gs.com
Goldman Sachs India SPL

Chris Kress

+1(212)902-6696 | chris.kress@gs.com
Goldman Sachs & Co. LLC

David Balaban

+1(212)357-6369 | david.balaban@gs.com
Goldman Sachs & Co. LLC

Key Data

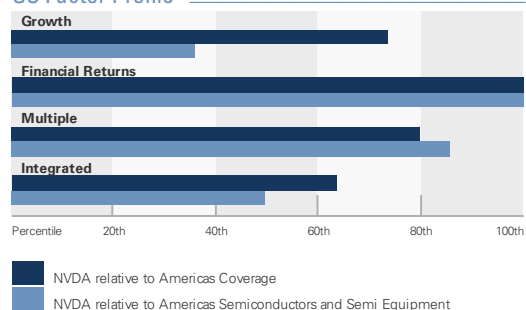
Market cap: \$3.1tr
Enterprise value: \$3.1tr
3m ADTV: \$42.5bn
United States
Americas Semiconductors and Semi Equipment
M&A Rank: 3
Americas Conviction List

GS Forecast

	1/24	1/25E	1/26E	1/27E
Revenue (\$ mn) New	60,922.0	128,598.0	185,894.0	214,248.5
Revenue (\$ mn) Old	60,922.0	127,556.5	173,486.6	203,423.1
EBITDA (\$ mn)	35,093.0	82,481.5	118,288.3	137,271.9
EBIT (\$ mn)	33,585.0	80,793.5	116,538.3	135,441.9
EPS (\$) New	1.15	2.74	4.06	4.86
EPS (\$) Old	1.15	2.77	3.94	4.85
P/E (X)	34.5	45.8	31.0	25.8
Dividend yield (%)	0.0	0.0	0.0	0.0
Net debt/EBITDA (X)	0.1	(0.3)	(0.5)	(0.7)

	7/24	10/24E	1/25E	4/25E
EPS (\$)	0.64	0.73	0.80	0.88

GS Factor Profile



Source: Company data, Goldman Sachs Research estimates.
See disclosures for details.

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Buy

CL

Nvidia Corp. (NVDA)

Rating since Feb 23, 2023

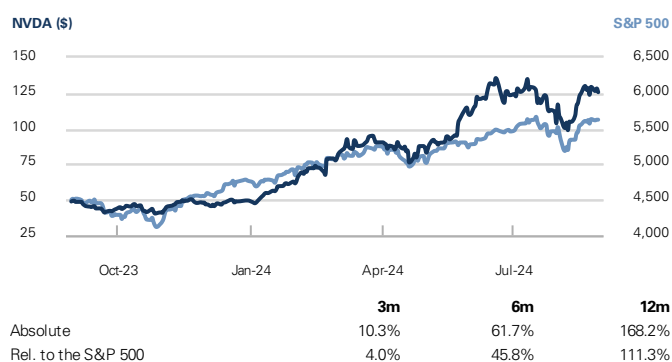
Ratios & Valuation

	1/24	1/25E	1/26E	1/27E
P/E (X)	34.5	45.8	31.0	25.8
EV/EBITDA (X)	28.0	37.1	25.2	20.9
EV/sales (X)	16.2	23.8	16.0	13.4
FCF yield (%)	2.8	2.1	3.2	4.1
EV/DACF (X)	30.7	42.2	28.2	23.4
CROCI (%)	66.4	108.0	129.0	137.3
ROE (%)	88.4	108.0	93.8	77.4
Net debt/EBITDA (X)	0.1	(0.3)	(0.5)	(0.7)
Net debt/equity (%)	5.6	(24.9)	(45.1)	(57.4)
Interest cover (X)	132.7	332.5	485.6	564.3
Inventory days	118.3	75.1	60.8	57.2
Receivable days	41.4	39.7	42.7	45.5
Days payable outstanding	44.1	41.1	35.3	33.8

Growth & Margins (%)

	1/24	1/25E	1/26E	1/27E
Total revenue growth	125.9	111.1	44.6	15.3
EBITDA growth	345.7	135.0	43.4	16.0
EPS growth	413.1	137.7	47.9	19.9
DPS growth	0.0	112.5	17.6	0.0
Gross margin	73.6	75.4	73.4	73.4
EBIT margin	55.1	62.8	62.7	63.2

Price Performance



Source: FactSet. Price as of 28 Aug 2024 close.

Income Statement (\$ mn)

	1/24	1/25E	1/26E	1/27E
Total revenue	60,922.0	128,598.0	185,894.0	214,248.5
Cost of goods sold	(16,105.0)	(31,614.4)	(49,431.8)	(56,948.3)
SG&A	(2,558.0)	(3,412.1)	(4,162.3)	(4,566.4)
R&D	(8,674.0)	(12,778.0)	(15,761.6)	(17,291.9)
Other operating inc./exp.)	—	—	—	—
EBITDA	35,093.0	82,481.5	118,288.3	137,271.9
Depreciation & amortization	(1,508.0)	(1,688.0)	(1,750.0)	(1,830.0)
EBIT	33,585.0	80,793.5	116,538.3	135,441.9
Net interest inc./exp.)	611.0	1,522.0	2,810.0	4,360.0
Income/(loss) from associates	—	—	—	—
Pre-tax profit	34,196.0	82,315.5	119,348.3	139,801.9
Provision for taxes	(5,433.0)	(14,274.4)	(20,289.2)	(23,766.3)
Minority interest	—	—	—	—
Preferred dividends	—	—	—	—
Net inc. (pre-exceptionals)	28,763.0	68,041.1	99,059.1	116,035.5
Net inc. (post-exceptionals)	29,759.0	69,496.1	99,059.1	116,035.5
EPS (basic, pre-exception) (\$)	1.17	2.77	4.10	4.92
EPS (diluted, pre-exception) (\$)	1.15	2.74	4.06	4.86
EPS (ex-ESO exp., dil.) (\$)	1.30	2.94	4.30	5.15
DPS (\$)	0.02	0.03	0.04	0.04
Div. payout ratio (%)	1.4	1.2	1.0	0.8
Wtd avg shares out. (basic) (mn)	24,692.5	24,545.2	24,168.5	23,593.9
Wtd avg shares out. (diluted) (mn)	24,932.5	24,807.5	24,425.8	23,854.7

Balance Sheet (\$ mn)

	1/24	1/25E	1/26E	1/27E
Cash & cash equivalents	7,281.0	29,083.9	66,286.5	106,849.0
Accounts receivable	9,999.0	17,957.2	25,505.7	27,868.4
Inventory	5,282.0	7,734.5	8,747.1	9,086.4
Other current assets	21,783.0	30,254.0	30,254.0	30,254.0
Total current assets	44,345.0	85,029.6	130,793.3	174,057.8
Net PP&E	3,914.0	5,040.0	5,290.0	5,460.0
Net intangibles	17,469.0	20,709.0	20,709.0	20,709.0
Total investments	0.0	0.0	0.0	0.0
Other long-term assets	—	—	—	—
Total assets	65,728.0	110,778.6	156,792.3	200,226.8
Accounts payable	2,699.0	4,416.8	5,148.0	5,398.2
Short-term debt	1,250.0	0.0	0.0	0.0
Current lease liabilities	—	—	—	—
Other current liabilities	6,682.0	10,289.0	10,289.0	10,289.0
Total current liabilities	10,631.0	14,705.8	15,437.0	15,687.2
Long-term debt	8,459.0	8,461.0	8,461.0	8,461.0
Non-current lease liabilities	—	—	—	—
Other long-term liabilities	3,660.0	4,640.0	4,640.0	4,640.0
Total long-term liabilities	12,119.0	13,101.0	13,101.0	13,101.0
Total liabilities	22,750.0	27,806.8	28,538.0	28,788.2
Preferred shares	—	—	—	—
Total common equity	42,978.0	82,971.8	128,254.3	171,438.6
Minority interest	—	—	—	—
Total liabilities & equity	65,728.0	110,778.6	156,792.3	200,226.8
BVPS (\$)	1.72	3.34	5.25	7.19

Cash Flow (\$ mn)

	1/24	1/25E	1/26E	1/27E
Net income	29,759.0	69,496.1	99,059.1	116,035.5
D&A add-back	1,508.0	1,688.0	1,750.0	1,830.0
Minority interest add-back	—	—	—	—
Net (inc)/dec working capital	(4,516.0)	(6,031.9)	(7,829.9)	(2,451.8)
Others	1,340.0	2,683.6	7,190.1	8,092.5
Cash flow from operations	28,091.0	67,835.7	100,169.3	123,506.3
Capital expenditures	(1,068.0)	(2,346.0)	(2,000.0)	(2,000.0)
Acquisitions	—	—	—	—
Divestitures	—	—	—	—
Others	(9,498.0)	(7,532.0)	—	—
Cash flow from investing	(10,566.0)	(9,878.0)	(2,000.0)	(2,000.0)
Dividends paid	(395.1)	(834.1)	(966.7)	(943.8)
Share issuance/(repurchase)	(9,533.0)	(30,898.0)	(60,000.0)	(80,000.0)
Inc/(dec) in debt	—	—	—	—
Others	(3,705.9)	(4,422.7)	—	—
Cash flow from financing	(13,634.0)	(36,154.8)	(60,966.7)	(80,943.8)
Total cash flow	3,891.0	21,802.9	37,202.6	40,562.5
Free cash flow	27,023.0	65,489.7	98,169.3	121,506.3
Free cash flow per share (basic) (\$)	1.09	2.67	4.06	5.15

Source: Company data, Goldman Sachs Research estimates.

mix of new products within Data Center. Non-GAAP EPS (excl. SBC) of \$0.68 was 6% above FactSet consensus.

Guidance: FY3Q (October) revenue guidance of \$32.5bn (+8% qoq, +79% yoy) at the mid-point came in 2% above Street consensus. Non-GAAP gross margin (excl. SBC) was guided to 75%, non-GAAP opex to \$3bn (+12% qoq), other income and expenses to ~\$350mn, and non-GAAP tax rate to 17%. Based on our calculation, this guidance implies a FY3Q non-GAAP EPS (excl. SBC) of ~\$0.73, which is 2% above consensus estimates heading into the print.

Exhibit 1: Actuals vs. Estimates

	FY2Q25A					FY3Q25E				
(\$mn except per share data)	Actual	GSe	Cons.	vs GSe	vs Cons.	Guidance	GSe	Cons.	vs GSe	vs Cons.
Revenue	30,040	29,769	28,737	1%	5%	32,500	34,216	31,714	-5%	2%
qoq	15%	14%	10%	--	--	8%	15%	10%	--	--
yoy	122%	120%	113%	--	--	79%	89%	75%	--	--
Segment revenue										
Gaming	2,880	2,925	2,708	-2%	6%					
qoq	9%	10%	2%	--	--					
yoy	16%	18%	9%	--	--					
Datacenter	26,272	25,947	25,270	1%	4%					
qoq	16%	15%	12%	--	--					
yoy	154%	151%	145%	--	--					
ProVis	454	457	450	-1%	1%					
qoq	6%	7%	5%	--	--					
yoy	20%	21%	19%	--	--					
Automotive	346	361	345	-4%	0%					
qoq	5%	10%	5%	--	--					
yoy	37%	43%	36%	--	--					
OEM & Other	88	79	78	12%	13%					
qoq	13%	1%	0%	--	--					
yoy	33%	19%	18%	--	--					
Gross profit	22,747	22,602	21,753	1%	5%					
Gross margin (ex ESO)	75.7%	75.9%	75.7%	-0.2%	0.0%	75.0%	75.2%	75.5%	-0.2%	-0.5%
Opex (ex ESO)	2,792	2,797	2,880	-0.2%	-3%	3,000	2,881	3,031	4.1%	-1.0%
Operating profit (ex ESO)	19,955	19,806	18,873	0.8%	6%	21,375	22,848	20,914		
% of revenue	66.4%	66.5%	65.7%	-0.1%	0.8%	65.8%	66.8%	65.9%		
EPS (non-GAAP ex ESO)	\$0.68	\$0.68	\$0.65	0%	6%	\$0.73	\$0.79	\$0.71	-8%	2%
qoq	11%	11%	5%	--	--	6%	16%	10%	--	--
yoy	153%	152%	139%	--	--	81%	97%	77%	--	--

Source: FactSet, Company data, Goldman Sachs Global Investment Research

Key Positives/Concerns and Implications for the Broader Group

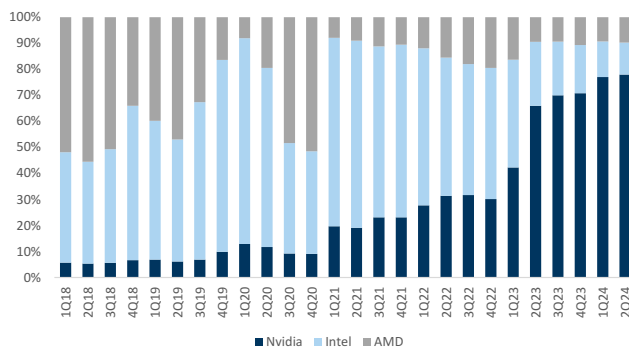
Positives: **1) Broad-based strength in Data Center:** the underlying demand environment in Data Center remains strong and broad-based across Nvidia's Cloud, Consumer Internet, and Enterprise customers. Consistent with FY1Q (April), Data Center revenue derived from the major Cloud Service Providers accounted for ~45% of segment revenue in FY2Q (July), while Consumer Internet and Enterprise customers combined for the remaining ~55%. **2) Blackwell to ramp in FY4Q (Jan):** following a re-design of the device, Nvidia is sampling Blackwell-based products broadly and expects several billion dollars in Blackwell revenue in FY4Q (January). We believe this outlook coupled with other bullish comments, particularly on customer reception, will remove arguably the most significant risk factor that had weighed on the stock ahead of the print, per our recent investor conversations. **3) Hopper strength:** despite the anticipation of Blackwell, customer demand for Hopper-based products, especially the H200 (i.e. >40% more memory bandwidth vs. the H100), remains strong with the company expecting revenue to grow hoh in FY2H and potentially qoq in FY4Q. **4) Doubling in Ethernet revenue:** following an unexpected sequential decline in FY1Q

driven by supply issues, Networking revenue grew 16% qoq in FY2Q with management citing a doubling of Ethernet for AI revenue. We believe this reinforces our view that, following the introduction of Spectrum-X, Nvidia is positioned to serve customers irrespective of the network communication protocol (i.e. InfiniBand vs. Ethernet). **5)**

FY2025 Sovereign AI revenue outlook increased: Nvidia increased its FY2025 Sovereign AI revenue outlook from high-single digit billions of dollars prior to low-double digit billions of dollars on today's earnings call. Beyond FY2025, we expect Sovereign AI revenue to grow at least in-line with overall Data Center revenue as sovereign states look to build AI models that are based on local datasets, language and/or culture. **6)**

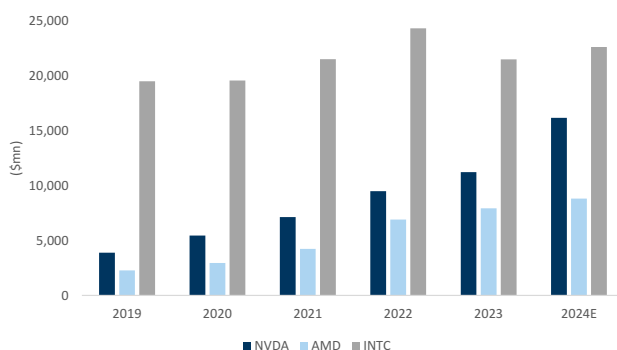
Steady growth in Software: while still a very small percentage of the overall business, management shared their expectation for total Software revenue (including software, SaaS and support revenue) to exit FY2025 at an annualized run-rate of \$2bn, up from ~\$1bn exiting FY2024, with Nvidia AI Enterprise being the notable contributor to growth. **7) Capital return:** supported by its robust FCF generation (\$13.5bn in FY2Q, \$47bn on a TTM basis), Nvidia returned \$7.4bn to shareholders in FY2Q (i.e. \$7.2bn in share repurchases + \$246mn in dividends) and, today, announced an additional \$50bn to their share repurchase authorization.

Exhibit 2: Nvidia's share of the Data Center Compute market has grown from ~15% five years ago to ~80% today
Data Center Compute Wallet Share



Source: Company data, Goldman Sachs Global Investment Research

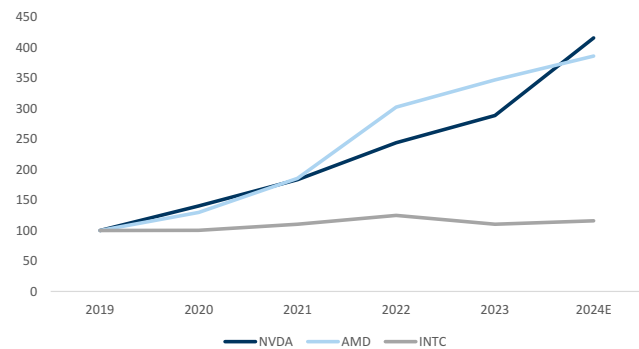
Exhibit 4: Nvidia's opex on an absolute basis is fast-approaching that of Intel
Annual opex (incl. SBC) - 2019-2024E



Source: Company data, Goldman Sachs Global Investment Research

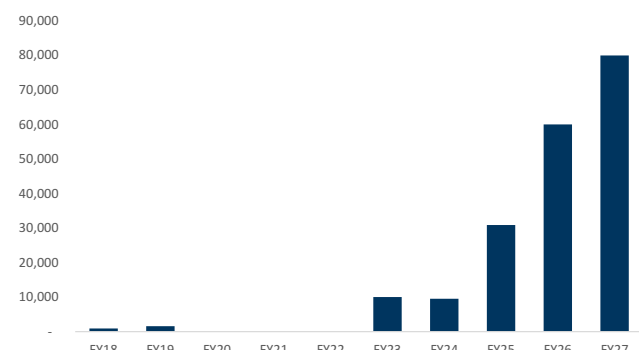
Exhibit 3: Nvidia's opex has increased by ~300% over the last 5 years

Annual opex (incl. SBC) indexed - 2019-2024E



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: We model a step-function increase in share repurchases
Share repurchases (\$mn)



Source: Company data, Goldman Sachs Global Investment Research

Concerns: **1) Sustainability of growth in AI infrastructure capex:** despite management's optimism that extends beyond the near-term, we expect the ongoing investor debate regarding the return profile on AI infrastructure investments and the implications this will likely have on future capital spending to persist, and for this to potentially cap NVDA's valuation multiple. **2) Supply uncertainty:** we believe key inputs including High-Bandwidth Memory (HBM) and advanced packaging (i.e. CoWoS) remain in tight supply. In fact, absent supply constraints, we believe the company's current backlog is such that Hopper revenue can grow sequentially in the January quarter. **3) Gross margin normalization:** management's full-year non-GAAP gross margin (excl. SBC) guidance of mid-70% (specifically, if one interprets this to be ~75%), implies another sequential decline in gross margins in FY4Q. While we believe our updated FY2026/27 non-GAAP gross margin forecast of 73.5% is achievable, we acknowledge that the fear of further gross margin degradation could weigh on the stock in the near-term. **4) Higher opex:** management now expects to grow full-year opex in the mid/high-40% range, up from its prior plan in the low-40%^s. While we tend to perceive upside to opex favorably (i.e. as a sign of a compelling/improving opportunity set), all else equal, it will mute Nvidia's rate of operating margin expansion.

Read-across: We believe the negative AH price action (-6.9%) in NVDA is a reaction to the company's gross margin outlook as opposed to Data Center market/demand dynamics. If anything, we believe Nvidia's comments on Data Center demand were constructive and bode well for others in our coverage universe that enable and are exposed to the build-out of AI infrastructure. In particular, we perceive ongoing tightness in High-Bandwidth Memory as a material positive for Buy-rated MU.

Estimate Update

We increase our FY2025/26/27 revenue estimates by 1%/7%/5%, respectively, on an improved Data Center outlook partially offset by more conservative estimates for Gaming and Automotive. We trim our FY2025 non-GAAP EPS (excl. SBC) estimate from \$2.96 to \$2.94 as lower gross margin and higher opex more than offset higher revenue. Our FY2026/27 non-GAAP EPS (excl. SBC) estimates, on the other hand, increase 3%/1%, respectively, on higher Data Center revenue partially offset by lower gross margins and higher opex.

Exhibit 6: We modestly increase FY2026-27 non-GAAP EPS (excl. SBC) estimates

NVDA New vs. Old (\$mn except per share data)	FY25			FY26			FY27		
	New	Old	% Change	New	Old	% Change	New	Old	% Change
Revenue	128,598	127,557	0.8%	185,894	173,487	7.2%	214,248	203,423	5.3%
Gross Margin (%)	75.6%	76.0%	-0.4%	73.5%	75.5%	-2.0%	73.5%	75.6%	-2.1%
Operating Margin (%)	66.6%	67.3%	-0.7%	65.9%	67.4%	-1.5%	66.4%	68.1%	-1.7%
EPS - Operating	\$2.94	\$2.96	-0.6%	\$4.30	\$4.16	3.4%	\$5.15	\$5.10	0.9%

Source: Goldman Sachs Global Investment Research

Bull/Bear Framework

Below we present a bull/bear framework to gauge NVDA's risk/reward profile at current levels. In short, we believe risk/reward on the stock is favorable with our most bullish scenario pointing to 97% potential upside vs. 60% potential downside under our most bearish scenario. See below for a brief summary of the assumptions supporting our

analysis.

Assumptions:

- In our most bearish scenario, we assume a 25% yoy decline in Data Center segment revenue tied to CSP customers, and a 50% yoy decline in Enterprise and Consumer Internet, in CY2025. We believe these assumptions are conservative enough to qualify as inputs for our “most bearish” scenario, especially given recent commentary from the Tier-1 CSPs indicating sustained capex growth through CY2025. All in, our updated Bear #2 scenario assumes \$71bn in Data Center segment revenue or 58% below our current central case.
- In our bearish scenario (“Bear #1”), we assume CY2025 Data Center segment revenue of \$119bn or the mid-point of our base case and most bearish scenario.
- In our most bullish scenario (“Bull #2”), we assume a yoy growth rate of 100% for the Data Center segment in CY2025 or \$226bn in revenue.
- In our bullish scenario (“Bull #1”), we assume CY2025 Data Center segment revenue of \$197bn or the mid-point of our base case and most bullish scenario.
- Gross margin assumptions for every business segment remain fixed across all scenarios (i.e. it is segment mix that influences corporate gross margin).
- We adjust opex and SBC in accordance with changes in revenue.
- All other line items (i.e. interest income and other, diluted share count) are unchanged from our base case assumptions.
- We assume target P/E multiples of 25x and 30x in our most bearish and bearish scenarios, and 35x and 40x in our bullish and most bullish scenarios. For context, as shown in [\(Exhibit 8\)](#) below, NVDA's P/E based on NTM consensus EPS has ranged between 20x and 65x since 2019 with a 43x median on a 3-year lookback.

Exhibit 7: Our Bull/Bear Analysis points to a favorable risk/reward profile for NVDA

Nvidia Bull/Bear Analysis based on CY2025 (\$mn, unless otherwise indicated)

	Bear # 2	Bear # 1	Base	Bull # 1	Bull # 2
Revenue	88,396	137,145	185,894	214,875	243,856
yoy %	-31%	7%	45%	67%	90%
Gross Profit	63,339	99,901	136,462	158,198	179,934
% of revenue	71.7%	72.8%	73.4%	73.6%	73.8%
Opex	16,265	17,937	19,924	21,273	22,191
yoy %	0%	11%	23%	31%	37%
Operating Income	47,074	81,963	116,538	136,925	157,743
% of revenue	53%	60%	63%	64%	65%
Interest & Other	2,810	2,810	2,810	2,810	2,810
Taxes	8,480	14,411	20,289	23,755	27,294
Tax rate	17.0%	17.0%	17.0%	17.0%	17.0%
Share Count	24,426	24,426	24,426	24,426	24,426
EPS (incl SBC)	\$1.70	\$2.88	\$4.06	\$4.75	\$5.46
EPS (excl SBC)	\$1.88	\$3.09	\$4.30	\$5.01	\$5.74
P/E Multiple*	25x	30x	31x	35x	40x
Valuation	\$47	\$93	\$135	\$175	\$230
vs. current price	-60%	-20%	15%	50%	97%

Segment revenue

Gaming	13,173	13,173	13,173	13,173	13,173
Pro-Viz	2,073	2,073	2,073	2,073	2,073
Data Center	70,664	119,412	168,161	197,142	226,123
yoy (%)	-38%	6%	49%	74%	100%
Automotive	2,099	2,099	2,099	2,099	2,099
OEM & Other	387	387	387	387	387

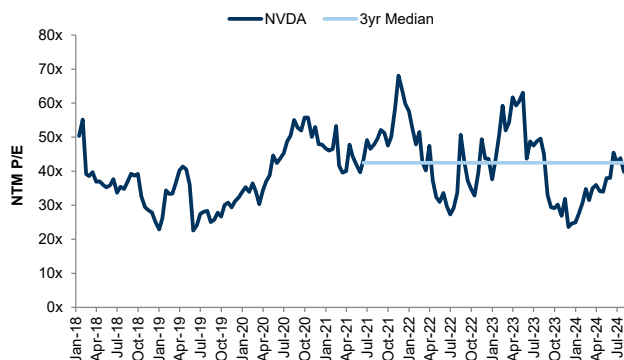
Segment gross margin

Gaming	58%	58%	58%	58%	58%
Pro-Viz	70%	70%	70%	70%	70%
Data Center	75%	75%	75%	75%	75%
Automotive	53%	53%	53%	53%	53%
OEM & Other	30%	30%	30%	30%	30%

*Base case multiple of 31x is implied from our current 12-month price target of \$135 which is based on 50x our normalized EPS estimate of \$2.70

Source: Goldman Sachs Global Investment Research

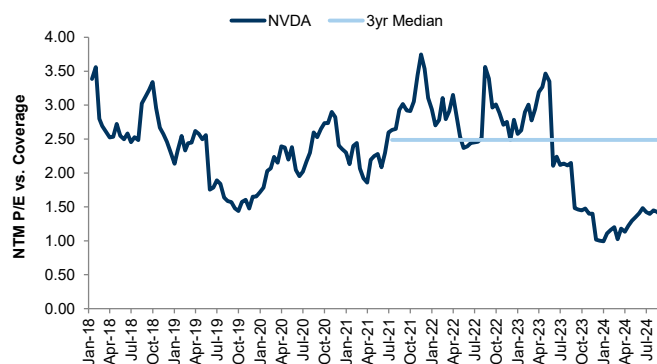
Exhibit 8: NVDA is trading at 39x or 9% below its 3-year median on NTM P/E



Source: FactSet

Exhibit 9: NVDA is trading on a relative NTM P/E multiple toward the low-end of its historical range

NVDA's NTM P/E vs. Coverage



Source: FactSet

Price Target Methodology and Key Risks

Our 12-month price target of \$135 (unchanged) is based on 50x (unchanged) our normalized EPS estimate of \$2.70 (unchanged). Note our price target of \$135 implies a 28x multiple on our updated Q5-Q8 non-GAAP EPS (excl. SBC) estimate of \$4.79. Key downside risks to our estimates and price target include: 1) a sudden decline in Gen AI infrastructure spend by the major CSPs and/or enterprises; 2) further restrictions on GPU exports; 3) weaker-than-expected demand for Gaming GPUs; 4) delays in new product introductions and their impact on revenue and profitability; and 5) supply chain issues.

Disclosure Appendix

Reg AC

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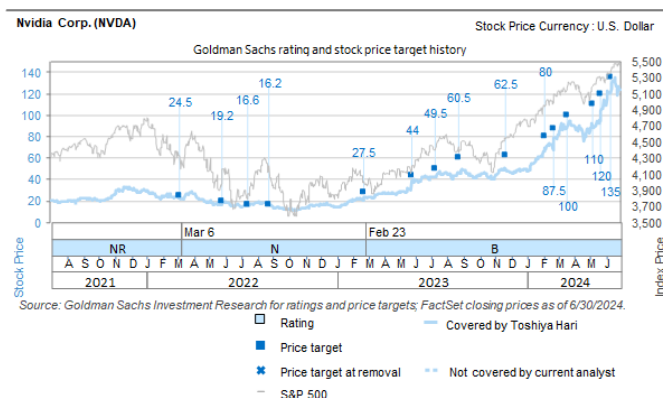
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